**Algorithm 1:** Powersort(A[1...n])

```
X := stack of runs ;                                   //capacity⌊lgn(n)⌋ + 1
P := stack of powers ;                                 //capacity ⌊lgn(n)⌋ + 1
s1 := 1; e1 = ExtendRunRight(A[1], n);
while e1 < n do
    s2 := e1 + 1; e2 := ExtendRunRight(A[s2], n) ;     //A[s2..e2] next run
    p := NodePower(s1, e1, s2, e2, n) ;   //Pj for node j between A[s1..e1]
     and A[s2..e2]
    while P.top() > p do
        ;              //previous merge deeper in tree than current
        P.pop() ;      //→ merge and replace run A[s1..e1] by result
        (s1, e1) := Merge(X.pop(), A[s1..e1]);
    end
    X.push(A[s1, e1]);
    P.push(p);
    s1 := s2; e1 := e2 ;              //Now A[s1..s1] is the rightmost run
end
while ¬X.empty() do
    (s1, e1) := Merge(X.pop(), A[s1..e1]);
end
```